Abstract

Using a simulated workforce dataset filtered to Data Science roles, we explore how AI adoption relates to productivity, innovation, hiring, revenue, and layoff risk. Principal component analysis reveals an AI-driven performance cluster distinct from workforce instability. Multiple linear regressions show AI adoption modestly raises revenue growth and innovation, but GDP growth and employee satisfaction are far stronger predictors. AI's link to layoff risk is small once human-centric factors are controlled. Random forest modelling offers no improvement over linear regression, indicating predominantly linear relationships. The findings portray AI as a performance enabler, not a wholesale replacement for talent, underscoring the need to balance technology adoption with human capital.

1. Loading the Required Libraries

```r
# Load essential libraries for data manipulation, visualization, and modeling
library(tidyverse)   # Collection of packages (dplyr, ggplot2, etc.)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.1
```

```
## Warning: package 'tibble' was built under R version 4.5.1
```

```
## Warning: package 'tidyr' was built under R version 4.5.1
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
## Warning: package 'purrr' was built under R version 4.5.2
```

```
## Warning: package 'dplyr' was built under R version 4.5.1
```

```
## Warning: package 'stringr' was built under R version 4.5.2
```

```
## Warning: package 'forcats' was built under R version 4.5.2
```

```
## Warning: package 'lubridate' was built under R version 4.5.1
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(janitor)     # Simple functions for cleaning data
```

```
## Warning: package 'janitor' was built under R version 4.5.3
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```r
library(corrplot)    # Correlation matrix visualization
```

```
## Warning: package 'corrplot' was built under R version 4.5.2
```

```
## corrplot 0.95 loaded
```

```r
library(factoextra)  # Extract and visualize PCA/eigenvalues
```

```
## Warning: package 'factoextra' was built under R version 4.5.2
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```r
library(caret)       # Classification and regression training
```

```
## Warning: package 'caret' was built under R version 4.5.2
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```r
library(randomForest)# Random forest algorithm
```

```
## Warning: package 'randomForest' was built under R version 4.5.2
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```r
library(Metrics)     # Evaluation metrics (e.g., RMSE)
```

```
## Warning: package 'Metrics' was built under R version 4.5.3

##
## Attaching package: 'Metrics'
##
## The following objects are masked from 'package:caret':
##
##     precision, recall
```

```r
library(psych)       # Descriptive statistics and factor analysis
```

```
## Warning: package 'psych' was built under R version 4.5.1

##
## Attaching package: 'psych'
##
## The following object is masked from 'package:randomForest':
##
##     outlier
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

2. Data Loading and Initial Inspection

```r
# Read the CSV file containing workforce transformation data
# Note: The file path is specific to the local machine; adjust as needed.
data <- read_csv("C:/Users/miraa/OneDrive/Documents/kaggle1/ai_workforce_transformation.csv")
```

```
## Rows: 1000000 Columns: 23
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (7): quarter, company, industry, region, department, economic_condition...
## dbl (16): year, inflation_rate, interest_rate, gdp_growth, ai_adoption_score...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Display the first 10 rows to understand the structure
head(data, n=10)
```

```
## # A tibble: 10 x 23
##     year quarter company     industry       region department economic_condition
##    <dbl> <chr>   <chr>       <chr>          <chr>  <chr>      <chr>
##  1  2011 Q1      DataNest    Data Analytics Europe Finance    Slowdown
```

```
## 2  2024 Q4      CyberWave   E-Commerce     Middl~ Engineeri~ AI Boom
## 3  2019 Q3      PixelCore   Gaming         Europe Product    Growth
## 4  2015 Q1      ApexAI      AI             Asia-~ Data Scie~ Growth
## 5  2012 Q1      TechNova    Software       Asia-~ Data Scie~ Slowdown
## 6  2025 Q2      FutureGrid  Cybersecurity  Middl~ Data Scie~ AI Boom
## 7  2011 Q1      ApexAI      AI             North~ Sales      Stable
## 8  2023 Q3      CloudSphere Cloud Computi~ North~ Sales      AI Boom
## 9  2015 Q4      ByteFusion  FinTech        Middl~ Finance    Growth
## 10  2015 Q4      ByteFusion  FinTech        South~ Operations Growth
## # i 16 more variables: work_model <chr>, inflation_rate <dbl>,
## #   interest_rate <dbl>, gdp_growth <dbl>, ai_adoption_score <dbl>,
## #   automation_level <dbl>, employees_hired <dbl>, layoffs <dbl>,
## #   attrition_rate <dbl>, employee_satisfaction <dbl>, avg_salary <dbl>,
## #   revenue_growth <dbl>, stock_growth <dbl>, productivity_index <dbl>,
## #   innovation_score <dbl>, layoff_risk_score <dbl>
```

2.1 Variable Descriptions

A brief description of each column heading (as provided by the dataset documentation):

year: Year of workforce and economic activity

quarter: Financial quarter (Q1, Q2, Q3, Q4)

company: Simulated company name

industry: Industry sector of the company

region: Geographic business region

department: Company department or business unit

economic_condition: Simulated macroeconomic environment

work_model: Work arrangement type (Remote, Hybrid, Onsite)

inflation_rate: Simulated inflation rate percentage

interest_rate: Simulated interest rate percentage

gdp_growth: Estimated GDP growth rate percentage

ai_adoption_score: Level of AI adoption within the organization

automation_level: Degree of business process automation

employees_hired: Number of employees hired during the period

layoffs: Number of employees laid off during the period

attrition_rate: Percentage of employees leaving voluntarily

employee_satisfaction: Overall employee satisfaction score

avg_salary: Average employee salary in USD

revenue_growth: Estimated company revenue growth percentage

stock_growth: Simulated stock market growth percentage

productivity_index: Overall workforce productivity score

innovation_score: Organizational innovation and technology advancement score

layoff_risk_score: Estimated workforce layoff risk score based on economic, operational, and workforce conditions

3. Data Preprocessing

3.1 Handling Missing Values

```r
# Count total missing values across the entire dataset
sum(is.na(data))
```

```
## [1] 30000
```

```r
# Remove rows containing any missing values (NA)
data1 <- data %>% drop_na()
```

3.2 Removing Duplicate Rows

```r
# Remove any duplicate rows to ensure unique records
data1 <- data1 %>% distinct()
```

3.3 Creating Numeric-Only Subset for Correlation

```r
# Select numeric variables for correlation analysis, dropping character columns
data2 <- select(data1, c("year", "inflation_rate":"layoff_risk_score"))

# Confirm the structure of the cleaned dataset
head(data1, n=10)
```

```
## # A tibble: 10 x 23
##     year quarter company     industry       region department economic_condition
##    <dbl> <chr>   <chr>       <chr>          <chr>  <chr>      <chr>
##  1  2011 Q1      DataNest    Data Analytics Europe Finance    Slowdown
##  2  2024 Q4      CyberWave   E-Commerce     Middl~ Engineeri~ AI Boom
##  3  2019 Q3      PixelCore   Gaming         Europe Product    Growth
##  4  2015 Q1      ApexAI      AI             Asia-~ Data Scie~ Growth
##  5  2012 Q1      TechNova    Software       Asia-~ Data Scie~ Slowdown
##  6  2025 Q2      FutureGrid  Cybersecurity  Middl~ Data Scie~ AI Boom
##  7  2011 Q1      ApexAI      AI             North~ Sales      Stable
##  8  2023 Q3      CloudSphere Cloud Computi~ North~ Sales      AI Boom
##  9  2015 Q4      ByteFusion  FinTech        Middl~ Finance    Growth
## 10  2015 Q4      ByteFusion  FinTech        South~ Operations Growth
## # i 16 more variables: work_model <chr>, inflation_rate <dbl>,
## #   interest_rate <dbl>, gdp_growth <dbl>, ai_adoption_score <dbl>,
## #   automation_level <dbl>, employees_hired <dbl>, layoffs <dbl>,
## #   attrition_rate <dbl>, employee_satisfaction <dbl>, avg_salary <dbl>,
## #   revenue_growth <dbl>, stock_growth <dbl>, productivity_index <dbl>,
## #   innovation_score <dbl>, layoff_risk_score <dbl>
```

```r
summary(data1)
```

```
##       year        quarter            company            industry        
##  Min.   :2005   Length:970299      Length:970299      Length:970299     
##  1st Qu.:2010   Class :character   Class :character   Class :character  
##  Median :2016   Mode  :character   Mode  :character   Mode  :character  
##  Mean   :2016                                                           
##  3rd Qu.:2021                                                           
##  Max.   :2026                                                           
##     region           department        economic_condition  work_model       
##  Length:970299      Length:970299      Length:970299      Length:970299     
##  Class :character   Class :character   Class :character   Class :character  
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character  
##                                                                             
##                                                                             
##                                                                             
##  inflation_rate   interest_rate      gdp_growth     ai_adoption_score
##  Min.   : 1.000   Min.   : 0.000   Min.   :-8.000   Min.   :  0.00   
##  1st Qu.: 2.650   1st Qu.: 2.490   1st Qu.: 1.310   1st Qu.: 13.00   
##  Median : 4.000   Median : 3.500   Median : 3.000   Median : 25.00   
##  Mean   : 4.059   Mean   : 3.507   Mean   : 2.997   Mean   : 28.98   
##  3rd Qu.: 5.350   3rd Qu.: 4.510   3rd Qu.: 4.680   3rd Qu.: 42.00   
##  Max.   :14.310   Max.   :10.800   Max.   :12.000   Max.   :100.00   
##  automation_level employees_hired    layoffs      attrition_rate 
##  Min.   :  0.00   Min.   :  100   Min.   :    0   Min.   : 2.00  
##  1st Qu.: 11.00   1st Qu.: 3071   1st Qu.:  849   1st Qu.:11.89  
##  Median : 25.00   Median : 6040   Median : 1699   Median :15.67  
##  Mean   : 29.11   Mean   : 6047   Mean   : 1971   Mean   :15.95  
##  3rd Qu.: 43.00   3rd Qu.: 9025   3rd Qu.: 2550   3rd Qu.:19.62  
##  Max.   :100.00   Max.   :11999   Max.   :16077   Max.   :45.00  
##  employee_satisfaction   avg_salary     revenue_growth     stock_growth    
##  Min.   :  1.00        Min.   : 30000   Min.   :-40.000   Min.   :-60.000  
##  1st Qu.: 57.20        1st Qu.: 57841   1st Qu.: -3.000   1st Qu.: -6.400  
##  Median : 65.60        Median : 78094   Median :  5.060   Median :  5.040  
##  Mean   : 65.07        Mean   : 80258   Mean   :  4.955   Mean   :  4.976  
##  3rd Qu.: 73.70        3rd Qu.:100737   3rd Qu.: 13.010   3rd Qu.: 16.420  
##  Max.   :100.00        Max.   :217594   Max.   : 70.250   Max.   : 90.930  
##  productivity_index innovation_score layoff_risk_score
##  Min.   :  6.30     Min.   :  1.00   Min.   :  0.00   
##  1st Qu.: 61.30     1st Qu.: 60.80   1st Qu.: 11.40   
##  Median : 68.70     Median : 71.50   Median : 32.10   
##  Mean   : 68.65     Mean   : 71.31   Mean   : 36.23   
##  3rd Qu.: 76.20     3rd Qu.: 82.30   3rd Qu.: 54.20   
##  Max.   :100.00     Max.   :100.00   Max.   :100.00   
```

4. Exploratory Data Analysis (EDA)

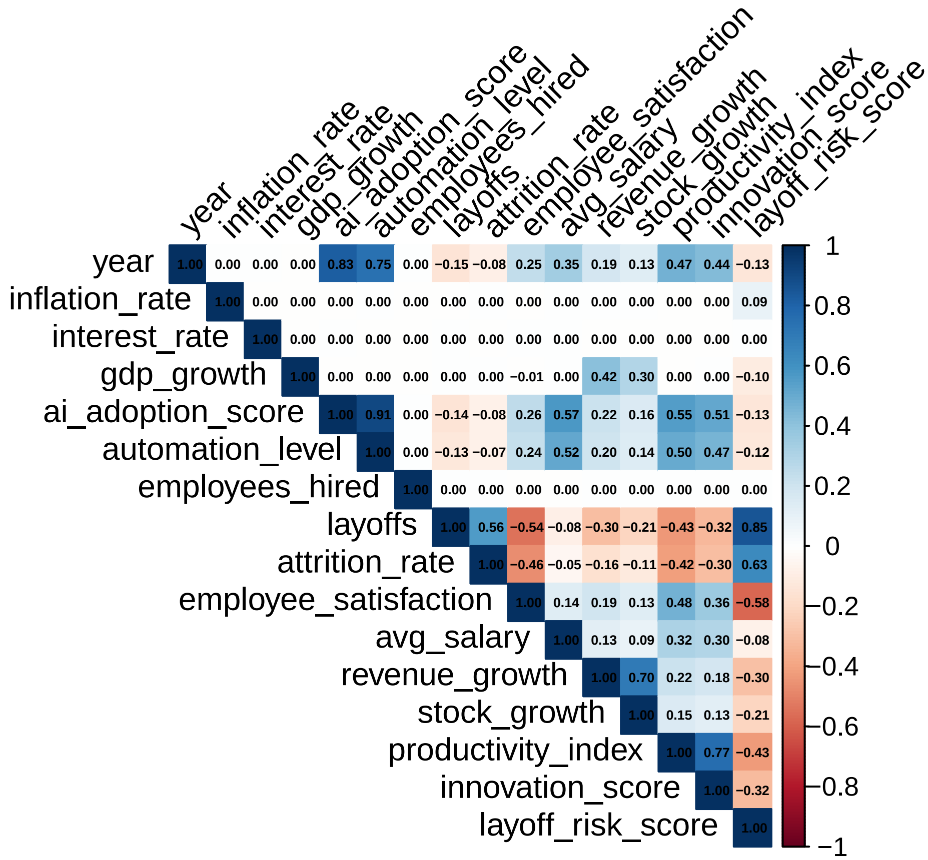
4.1 Filtering for Data Science Department

```r
# Focus on the Data Science department to evaluate AI impact on business analysis
data1.1 <- filter(data1, department == "Data Science")
data1.2 <- select(data1.1, c("year", "inflation_rate":"layoff_risk_score"))

# Quick look at the filtered data
head(data1.1, n=10)
```

```
## # A tibble: 10 x 23
##     year quarter company           industry region department economic_condition
##    <dbl> <chr>   <chr>             <chr>    <chr>  <chr>      <chr>
##  1  2015 Q1      ApexAI            AI       Asia-~ Data Scie~ Growth
##  2  2012 Q1      TechNova          Software Asia-~ Data Scie~ Slowdown
##  3  2025 Q2      FutureGrid        Cyberse~ Middl~ Data Scie~ AI Boom
##  4  2016 Q4      CloudSphere       Cloud C~ South~ Data Scie~ Slowdown
##  5  2006 Q3      CyberWave         E-Comme~ Europe Data Scie~ Slowdown
##  6  2020 Q1      ApexAI            AI       Europe Data Scie~ Pandemic
##  7  2019 Q4      QuantumEdge       AI       North~ Data Scie~ Growth
##  8  2007 Q4      FutureGrid        Cyberse~ Middl~ Data Scie~ Slowdown
##  9  2023 Q4      ApexAI            AI       Middl~ Data Scie~ AI Boom
## 10  2013 Q4      NeuroLink Systems AI       South~ Data Scie~ Growth
## # i 16 more variables: work_model <chr>, inflation_rate <dbl>,
## #   interest_rate <dbl>, gdp_growth <dbl>, ai_adoption_score <dbl>,
## #   automation_level <dbl>, employees_hired <dbl>, layoffs <dbl>,
## #   attrition_rate <dbl>, employee_satisfaction <dbl>, avg_salary <dbl>,
## #   revenue_growth <dbl>, stock_growth <dbl>, productivity_index <dbl>,
## #   innovation_score <dbl>, layoff_risk_score <dbl>
```

4.2 Correlation Matrix for Data Science Numeric Variables

```r
# Compute pairwise correlations using available observations
correlation <- cor(data1.2, use = 'pairwise.complete.obs')

# Visualize the correlation matrix with coefficients
corrplot(correlation, method = "color", type = "upper",
         tl.col = "black", addCoef.col = "black", number.cex = 0.45,
         tl.cex = 1, tl.srt = 45)
```



4.3 Unique Values in Character Columns

```r
# Explore the distinct categories in key categorical variables
unique(data1$quarter)
```

```
## [1] "Q1" "Q4" "Q3" "Q2"
```

```r
unique(data1$company)
```

```
##  [1] "DataNest"          "CyberWave"         "PixelCore"
##  [4] "ApexAI"            "TechNova"          "FutureGrid"
##  [7] "CloudSphere"       "ByteFusion"        "NeuroLink Systems"
## [10] "QuantumEdge"
```

```r
unique(data1$industry)
```

```
## [1] "Data Analytics"  "E-Commerce"      "Gaming"          "AI"
## [5] "Software"        "Cybersecurity"   "Cloud Computing" "FinTech"
```

```r
unique(data1$region)
```

```
## [1] "Europe"        "Middle East"   "Asia-Pacific"  "North America"
## [5] "South America"
```

```r
unique(data1$department)
```

```
## [1] "Finance"      "Engineering"  "Product"      "Data Science" "Sales"
## [6] "Operations"   "HR"           "Marketing"
```

```r
unique(data1$economic_condition)
```

```
## [1] "Slowdown"  "AI Boom"   "Growth"    "Stable"    "Recession" "Pandemic"
```

```r
unique(data1$work_model)
```

```
## [1] "Onsite" "Remote" "Hybrid"
```

5. Impact of AI on Data Science Department

To evaluate the impact of artificial intelligence on business analysis, we filter the dataset for the Data Science department from each company.

```r
data3 <- filter(data1, department == "Data Science")
head(data3)
```

```
## # A tibble: 6 x 23
##    year quarter company industry region department economic_condition work_model
##   <dbl> <chr>   <chr>   <chr>    <chr>  <chr>      <chr>              <chr>
## 1  2015 Q1      ApexAI  AI       Asia-~ Data Scie~ Growth             Onsite
## 2  2012 Q1      TechNo~ Software Asia-~ Data Scie~ Slowdown           Onsite
## 3  2025 Q2      Future~ Cyberse~ Middl~ Data Scie~ AI Boom            Remote
## 4  2016 Q4      CloudS~ Cloud C~ South~ Data Scie~ Slowdown           Hybrid
## 5  2006 Q3      CyberW~ E-Comme~ Europe Data Scie~ Slowdown           Hybrid
## 6  2020 Q1      ApexAI  AI       Europe Data Scie~ Pandemic           Remote
## # i 15 more variables: inflation_rate <dbl>, interest_rate <dbl>,
## #   gdp_growth <dbl>, ai_adoption_score <dbl>, automation_level <dbl>,
## #   employees_hired <dbl>, layoffs <dbl>, attrition_rate <dbl>,
## #   employee_satisfaction <dbl>, avg_salary <dbl>, revenue_growth <dbl>,
## #   stock_growth <dbl>, productivity_index <dbl>, innovation_score <dbl>,
## #   layoff_risk_score <dbl>
```

```r
describe(data3)   # psych::describe() gives descriptive statistics
```

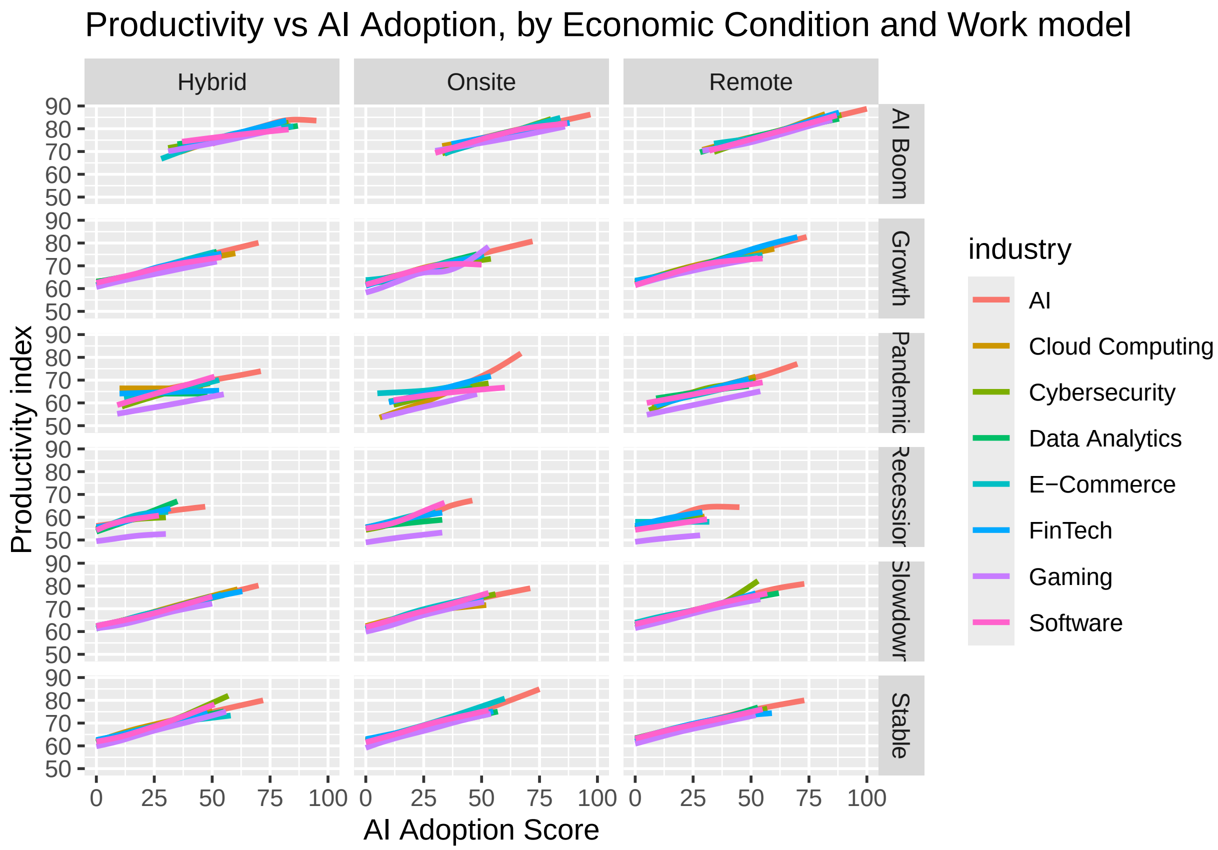
```
##                        vars      n    mean      sd  median trimmed     mad
## year                      1 120818 2015.50    6.36 2016.00 2015.50    8.90
## quarter*                  2 120818    2.50    1.12    3.00    2.50    1.48
## company*                  3 120818    5.49    2.87    5.00    5.49    2.97
## industry*                 4 120818    3.80    2.48    4.00    3.63    2.97
## region*                   5 120818    3.00    1.41    3.00    3.00    1.48
## department*               6 120818    1.00    0.00    1.00    1.00    0.00
```

```
## economic_condition*      7 120818     3.63     1.90     4.00     3.66     2.97
## work_model*              8 120818     1.99     0.82     2.00     1.99     1.48
## inflation_rate           9 120818     4.05     1.88     4.00     4.00     1.99
## interest_rate           10 120818     3.50     1.49     3.50     3.50     1.50
## gdp_growth              11 120818     2.99     2.50     2.99     2.99     2.51
## ai_adoption_score       12 120818    29.01    20.97    25.00    27.35    20.76
## automation_level        13 120818    29.13    22.22    26.00    27.24    23.72
## employees_hired         14 120818  6064.05  3432.57  6061.00  6066.34  4409.25
## layoffs                 15 120818  1976.23  1684.95  1701.00  1707.26  1255.76
## attrition_rate          16 120818    15.96     6.01    15.66    15.76     5.75
## employee_satisfaction   17 120818    65.07    12.88    65.60    65.44    12.31
## avg_salary              18 120818 80264.68 29778.69 78150.00 79143.61 31631.27
## revenue_growth          19 120818     4.90    11.94     4.99     4.96    11.88
## stock_growth            20 120818     4.91    16.92     4.91     4.92    16.87
## productivity_index      21 120818    68.66    11.21    68.70    68.72    11.12
## innovation_score        22 120818    71.30    15.50    71.40    71.52    16.01
## layoff_risk_score       23 120818    36.28    29.35    32.30    33.25    31.58
##                           min       max     range  skew kurtosis    se
## year                   2005.0   2026.00     21.00  0.00    -1.21  0.02
## quarter*                  1.0      4.00      3.00  0.00    -1.35  0.00
## company*                  1.0     10.00      9.00  0.00    -1.22  0.01
## industry*                 1.0      8.00      7.00  0.31    -1.32  0.01
## region*                   1.0      5.00      4.00  0.00    -1.30  0.00
## department*               1.0      1.00      0.00   NaN      NaN  0.00
## economic_condition*       1.0      6.00      5.00 -0.11    -1.58  0.01
## work_model*               1.0      3.00      2.00  0.01    -1.50  0.00
## inflation_rate            1.0     13.45     12.45  0.30    -0.35  0.01
## interest_rate             0.0     10.17     10.17  0.06    -0.14  0.00
## gdp_growth               -8.0     12.00     20.00  0.00     0.00  0.01
## ai_adoption_score         0.0    100.00    100.00  0.65    -0.33  0.06
## automation_level          0.0    100.00    100.00  0.66    -0.26  0.06
## employees_hired         100.0  11999.00  11899.00  0.00    -1.20  9.88
## layoffs                   0.0  15931.00  15931.00  2.22     7.36  4.85
## attrition_rate            2.0     45.00     43.00  0.42     0.63  0.02
## employee_satisfaction     1.0    100.00     99.00 -0.36     0.60  0.04
## avg_salary            30000.0 197243.00 167243.00  0.35    -0.39 85.67
## revenue_growth          -40.0     55.54     95.54 -0.05     0.04  0.03
## stock_growth            -60.0     76.46    136.46 -0.01     0.00  0.05
## productivity_index        9.7    100.00     90.30 -0.09     0.12  0.03
## innovation_score          1.0    100.00     99.00 -0.18    -0.27  0.04
## layoff_risk_score         0.0    100.00    100.00  0.65    -0.43  0.08
```

5.1 Productivity vs AI Adoption (by Economic Condition and Work Model)

```r
plot5 <- ggplot(data = data3) +
  geom_smooth(mapping = aes(x = ai_adoption_score, y = productivity_index, colour = industry),
              se = FALSE) +
  facet_grid(economic_condition ~ work_model) +
  labs(x = "AI Adoption Score", y = "Productivity index",
       title = "Productivity vs AI Adoption, by Economic Condition and Work model")
plot5
```

```
## 'geom_smooth()' using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

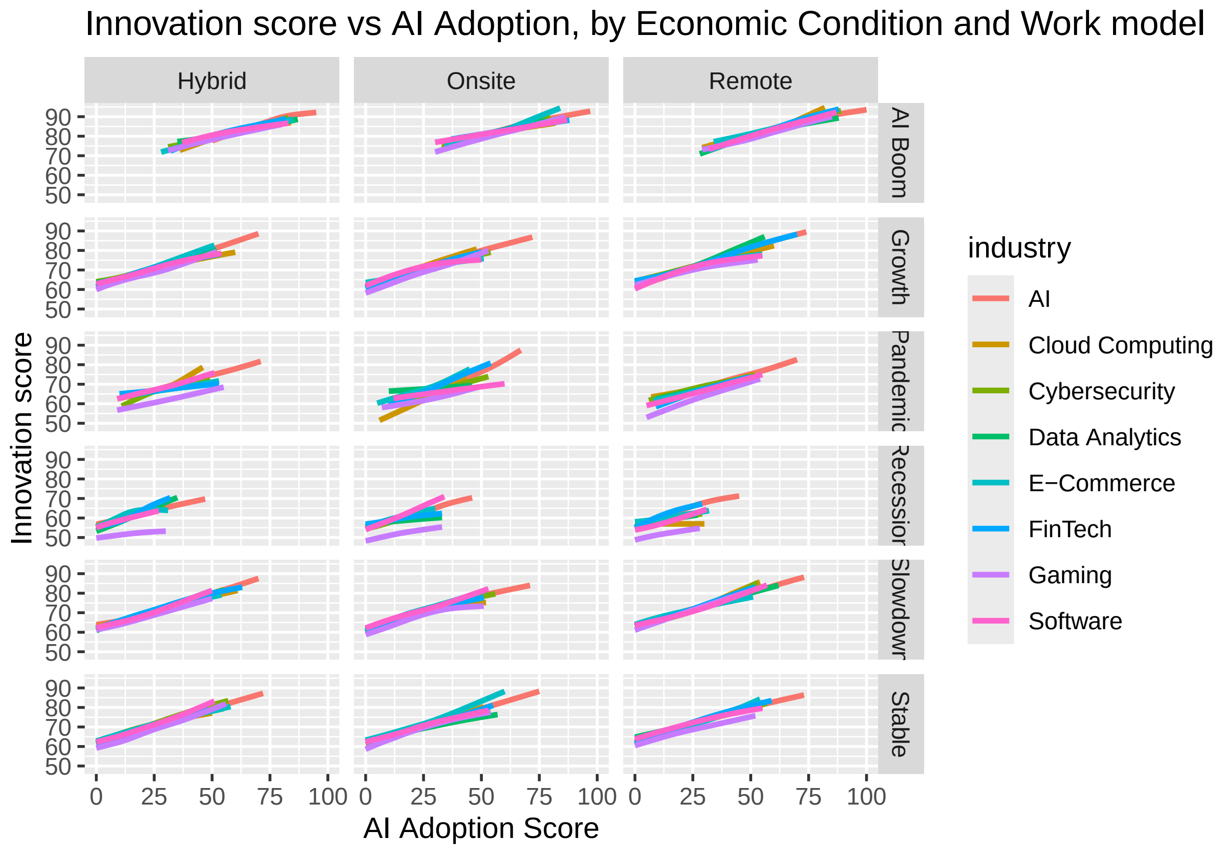


Observations: Productivity increased noticeably with AI adoption in the AI industry during the pandemic for onsite workers. In contrast, productivity plateaued for remote workers with increasing AI adoption during recession, and for hybrid workers during the AI boom.

5.2 Innovation Score vs AI Adoption

```r
plot6 <- ggplot(data = data3) +
  geom_smooth(mapping = aes(x = ai_adoption_score, y = innovation_score, colour = industry),
              se = FALSE) +
  facet_grid(economic_condition ~ work_model) +
  labs(x = "AI Adoption Score", y = "Innovation score",
       title = "Innovation score vs AI Adoption, by Economic Condition and Work model")
plot6
```

```
## 'geom_smooth()' using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



Observations: The gaming industry appears to have lower productivity (innovation score) with increased AI adoption compared to other industries during recession, irrespective of the work model.

5.3 Employees Hired vs AI Adoption

```
plot7 <- ggplot(data = data3) +
  geom_smooth(mapping = aes(x = ai_adoption_score, y = employees_hired, colour = industry),
              se = FALSE) +
  facet_grid(economic_condition ~ work_model) +
  labs(x = "AI Adoption Score", y = "Employees hired",
       title = "Employees hired vs AI Adoption, by Economic Condition and Work model")
plot7
```

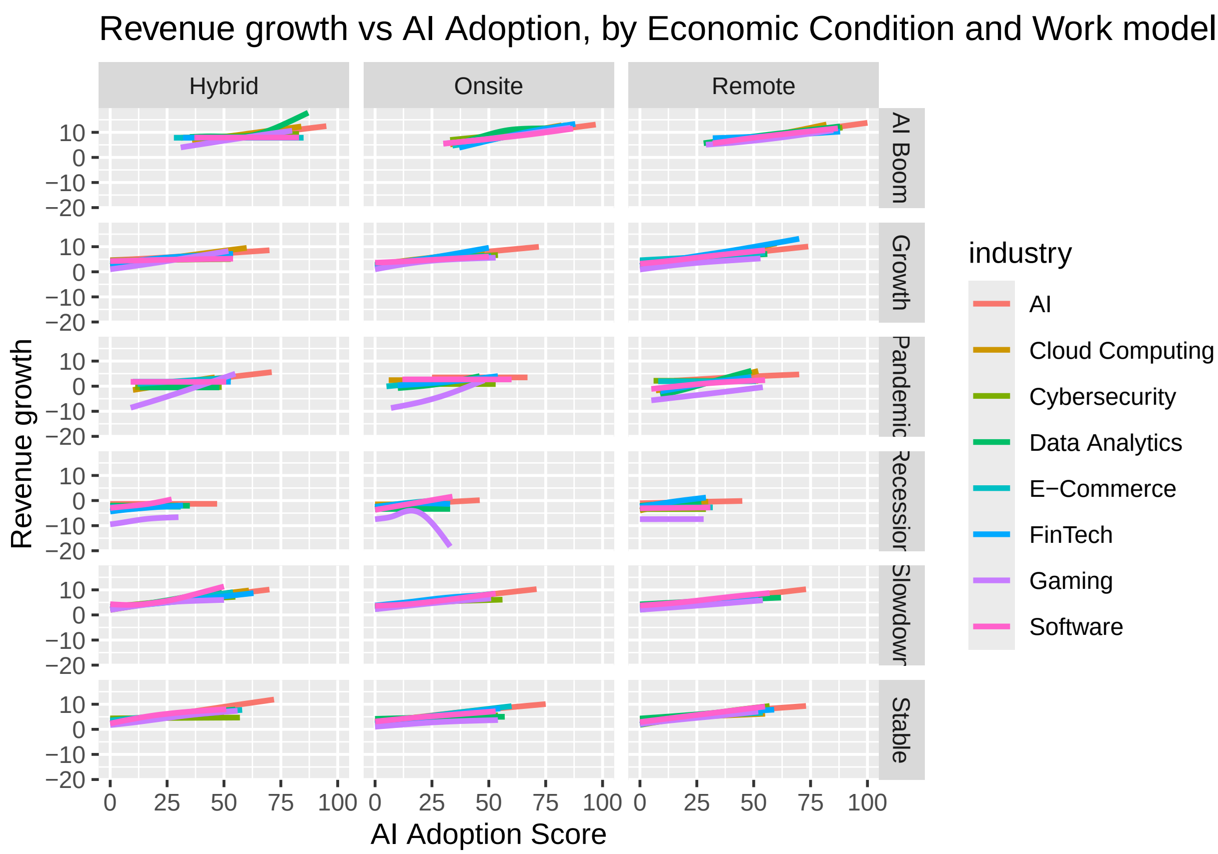
```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

Observations: The cybersecurity industry experienced a sharp but brief reduction in onsite hiring during the AI boom. Similarly, during the economic growth phase, employee recruitment decreased with increasing AI adoption for onsite workers, followed by a stabilization and then a further decline.

5.4 Revenue Growth vs AI Adoption

```r
plot8 <- ggplot(data = data3) +
  geom_smooth(mapping = aes(x = ai_adoption_score, y = revenue_growth, colour = industry),
              se = FALSE) +
  facet_grid(economic_condition ~ work_model) +
  labs(x = "AI Adoption Score", y = "Revenue growth",
       title = "Revenue growth vs AI Adoption, by Economic Condition and Work model")
plot8
```

```
## 'geom_smooth()' using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

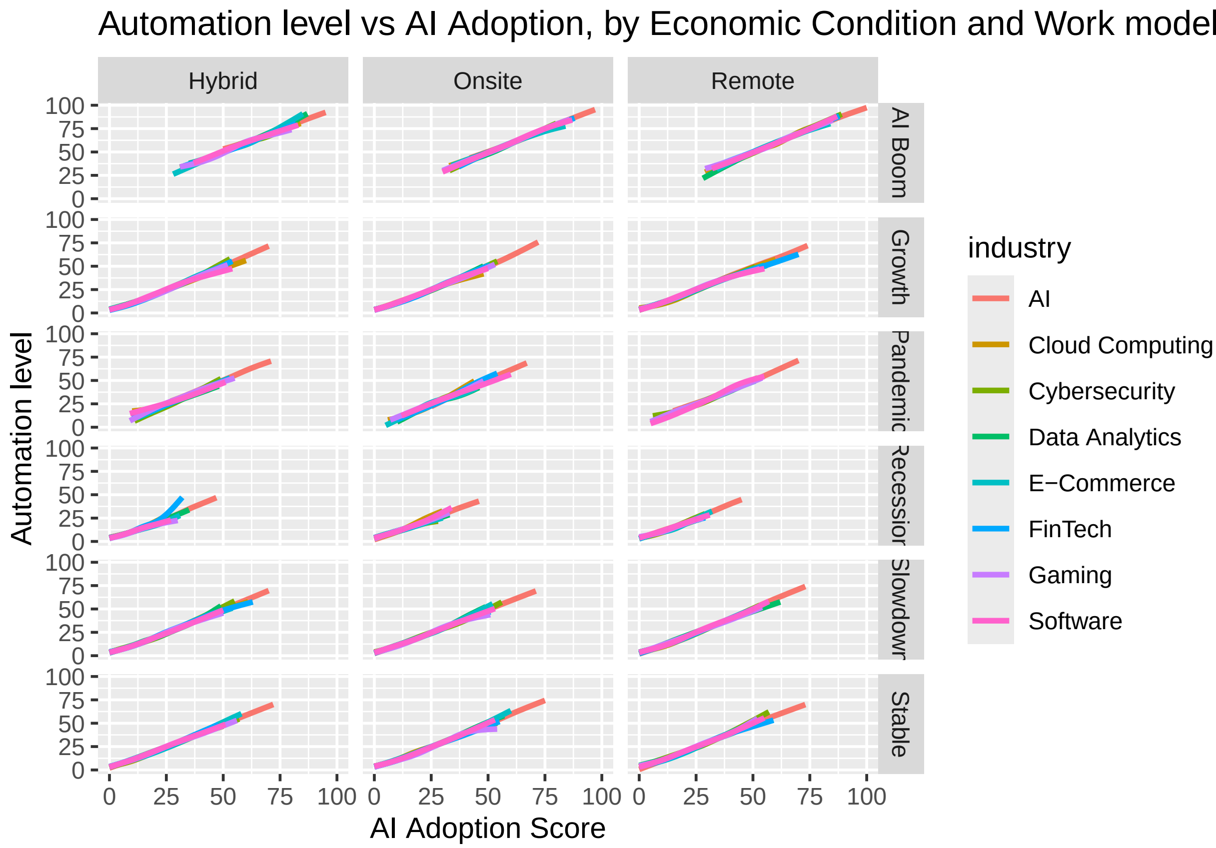


Observations: Revenue growth dropped in the gaming industry with increased AI adoption, especially pronounced for onsite workers during recession. However, during the pandemic, the gaming industry recovered revenue growth compared to other industries with an increase in AI adoption across all work models.

5.5 Automation Level vs AI Adoption

```
plot9 <- ggplot(data = data3) +
  geom_smooth(mapping = aes(x = ai_adoption_score, y = automation_level, colour = industry),
              se = FALSE) +
  facet_grid(economic_condition ~ work_model) +
  labs(x = "AI Adoption Score", y = "Automation level",
       title = "Automation level vs AI Adoption, by Economic Condition and Work model")
plot9
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



Observations: A marked increase in automation level with AI adoption was observed in the later part of the curve for the FinTech industry, particularly for hybrid workers during recession.

To determine whether differences between industries across economic conditions and work models are statistically significant, ANOVA followed by Tukey's HSD could be conducted. However, given the synthetic nature of the dataset, strong conclusions cannot be drawn.

6. Principal Component Analysis (PCA)

6.1 Preparing Data for PCA

```r
# Select only numeric variables from the Data Science subset
data4 <- select(data3, "year", "inflation_rate":"layoff_risk_score")

# Scale the data to have mean=0 and variance=1 (necessary for PCA)
data5 <- scale(data4, center = TRUE, scale = TRUE)
```

6.2 Performing PCA

```r
# Perform PCA on the scaled data; already centered and scaled
pca_result <- prcomp(data5, center = FALSE, scale. = FALSE)
```

```r
# View summary to decide how many principal components to retain
summary(pca_result)
```

```
## Importance of components:
##                           PC1    PC2    PC3     PC4     PC5     PC6     PC7
## Standard deviation     2.1733 1.5605 1.3309 1.00417 1.00068 0.99903 0.88448
## Proportion of Variance 0.2952 0.1522 0.1107 0.06302 0.06258 0.06238 0.04889
## Cumulative Proportion  0.2952 0.4474 0.5581 0.62112 0.68371 0.74609 0.79498
##                           PC8     PC9    PC10   PC11    PC12    PC13    PC14
## Standard deviation     0.8485 0.82237 0.71864 0.6800 0.51969 0.47870 0.45985
## Proportion of Variance 0.0450 0.04227 0.03228 0.0289 0.01688 0.01432 0.01322
## Cumulative Proportion  0.8400 0.88225 0.91453 0.9434 0.96031 0.97463 0.98785
##                          PC15    PC16
## Standard deviation     0.3534 0.26377
## Proportion of Variance 0.0078 0.00435
## Cumulative Proportion  0.9957 1.00000
```

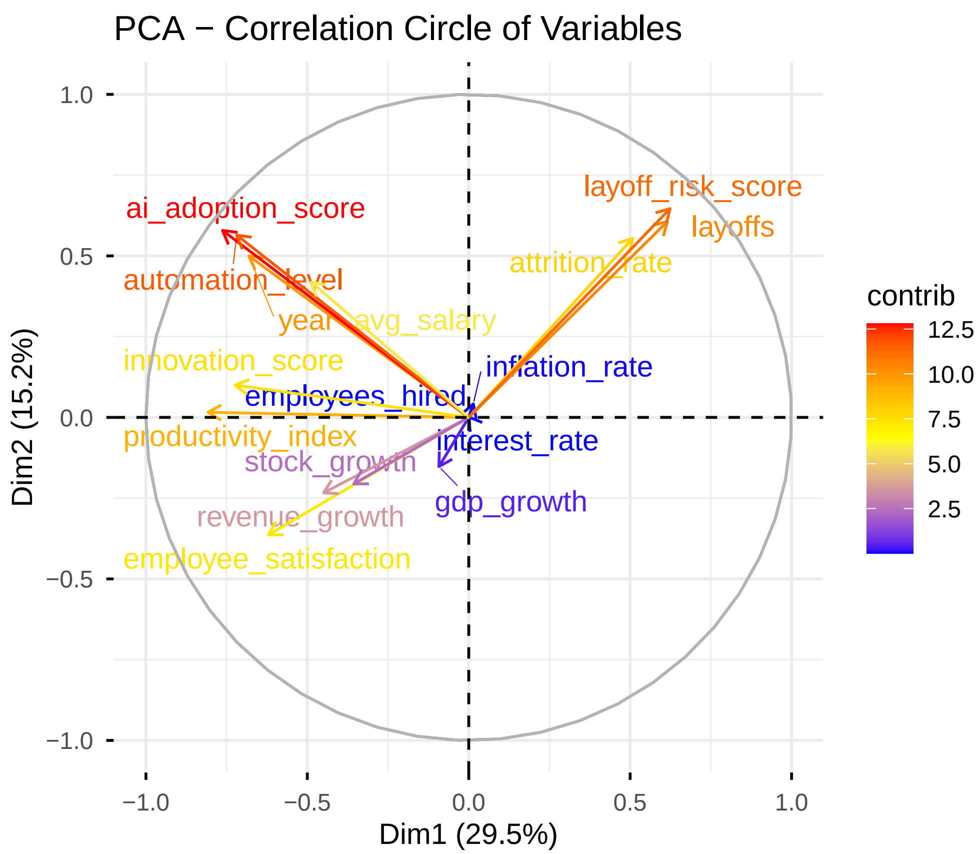
6.3 Visualizing Variable Contributions

```r
# Plot correlation circle showing variable contributions to PC1 and PC2
plot10 <- fviz_pca_var(pca_result,
                       col.var = "contrib",          # Color by contribution
                       gradient.cols = c("blue", "yellow", "red"),
                       repel = TRUE,                 # Prevent label overlap
                       title = "PCA - Correlation Circle of Variables")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## i The deprecated feature was likely used in the ggpubr package.
##   Please report the issue at <https://github.com/kassambara/ggpubr/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the factoextra package.
##   Please report the issue at <https://github.com/kassambara/factoextra/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
plot10
```



PCA Findings: PCA revealed two dominant dimensions explaining 44.7% of the total variance. The first component captured an AI-driven organizational performance dimension: higher AI adoption, automation, innovation, productivity, employee satisfaction, and revenue growth all loaded positively. On the opposite side, layoff risk, layoffs, and attrition clustered together, representing workforce instability. The close proximity of AI adoption and automation suggests that digital transformation initiatives are strongly associated with process automation and innovation.

7. Multiple Linear Regression

7.1 Revenue Growth Prediction

```
# Build linear model to predict revenue growth
model_rev <- lm(
  revenue_growth ~
    ai_adoption_score +
    automation_level +
    productivity_index +
    innovation_score +
    employee_satisfaction +
    gdp_growth +
    inflation_rate,
  data = data4
)
```

```r
summary(model_rev)
```

```
##
## Call:
## lm(formula = revenue_growth ~ ai_adoption_score + automation_level +
##     productivity_index + innovation_score + employee_satisfaction +
##     gdp_growth + inflation_rate, data = data4)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -46.678  -6.948   0.031   6.998  40.664
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           -16.672272   0.223596 -74.564   <2e-16 ***
## ai_adoption_score       0.079786   0.003521  22.660   <2e-16 ***
## automation_level        0.001920   0.003192   0.602    0.547
## productivity_index      0.085496   0.004579  18.671   <2e-16 ***
## innovation_score        0.004397   0.003053   1.440    0.150
## employee_satisfaction   0.107270   0.002629  40.806   <2e-16 ***
## gdp_growth              2.007308   0.011907 168.582   <2e-16 ***
## inflation_rate          0.009184   0.015809   0.581    0.561
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.35 on 120810 degrees of freedom
## Multiple R-squared:  0.2483, Adjusted R-squared:  0.2483
## F-statistic:  5702 on 7 and 120810 DF,  p-value: < 2.2e-16
```

Interpretation: For every 1-unit increase in AI adoption score, revenue growth increases by 0.0798 units, holding other variables constant.

```r
summary(model_rev)$r.squared
```

```
## [1] 0.2483238
```

The model explains approximately 24.8% of the variation in revenue growth ($R^2 = 0.248$, $p < 0.001$). AI adoption, productivity, employee satisfaction, and GDP growth were significant positive predictors. GDP growth exhibited the strongest effect. Automation level, innovation score, and inflation rate were not significant predictors after controlling for other variables.

7.2 Layoff Risk Score Prediction

```r
# Predict layoff risk score using organizational and workforce metrics
layoff_model <- lm(
  layoff_risk_score ~
    ai_adoption_score +
    automation_level +
    revenue_growth +
```

```r
    productivity_index +
    employee_satisfaction +
    attrition_rate,
  data = data1
)

summary(layoff_model)
```

```
##
## Call:
## lm(formula = layoff_risk_score ~ ai_adoption_score + automation_level +
##     revenue_growth + productivity_index + employee_satisfaction +
##     attrition_rate, data = data1)
##
## Residuals:
##    Min      1Q Median     3Q    Max
## -85.82 -14.24  -1.00  13.27  90.79
##
## Coefficients:
##                        Estimate Std. Error  t value Pr(>|t|)
## (Intercept)           69.656178   0.211417  329.472   <2e-16 ***
## ai_adoption_score      0.106799   0.002413   44.268   <2e-16 ***
## automation_level       0.003804   0.002177    1.748   0.0805 .
## revenue_growth        -0.404403   0.001775 -227.828   <2e-16 ***
## productivity_index    -0.280533   0.002516 -111.478   <2e-16 ***
## employee_satisfaction -0.738554   0.001915 -385.635   <2e-16 ***
## attrition_rate         2.049761   0.004058  505.107   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.03 on 970292 degrees of freedom
## Multiple R-squared:  0.5349, Adjusted R-squared:  0.5349
## F-statistic: 1.86e+05 on 6 and 970292 DF,  p-value: < 2.2e-16
```

Interpretation:

The set of predictors collectively has a significant effect ($p < 0.001$).

The model explains approximately 53.5% of the variation in layoff risk score ($R^2 = 0.535$), indicating a good fit according to typical business analytics thresholds (<30% weak, 30-50% moderate, 50-70% good).

A 1-point increase in AI adoption score increases layoff risk by 0.107 points, suggesting a modest positive association.

Revenue growth, employee satisfaction, and productivity significantly reduce layoff risk, with attrition rate being the strongest positive driver.

Automation level was not a significant predictor after controlling for other factors.

7.3 Innovation Score Prediction

```r
# Predict innovation score
model_innova <- lm(
```

```r
  innovation_score ~
    ai_adoption_score +
    automation_level +
    productivity_index +
    revenue_growth +
    employee_satisfaction +
    gdp_growth +
    inflation_rate,
  data = data4
)

summary(model_innova)
```

```
##
## Call:
## lm(formula = innovation_score ~ ai_adoption_score + automation_level +
##     productivity_index + revenue_growth + employee_satisfaction +
##     gdp_growth + inflation_rate, data = data4)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -43.816  -6.564   0.104   6.706  37.236
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            2.2308793  0.2153939  10.357   <2e-16 ***
## ai_adoption_score      0.0927614  0.0033141  27.990   <2e-16 ***
## automation_level       0.0003987  0.0030081   0.133    0.895
## productivity_index     0.9679724  0.0033039 292.981   <2e-16 ***
## revenue_growth         0.0039046  0.0027110   1.440    0.150
## employee_satisfaction -0.0017901  0.0024941  -0.718    0.473
## gdp_growth             0.0089289  0.0124701   0.716    0.474
## inflation_rate        -0.0049857  0.0148971  -0.335    0.738
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.757 on 120810 degrees of freedom
## Multiple R-squared:  0.6037, Adjusted R-squared:  0.6037
## F-statistic: 2.629e+04 on 7 and 120810 DF,  p-value: < 2.2e-16
```

Interpretation: The model explains 60.4% of the variation in innovation score ($R^2 = 0.604$). AI adoption and productivity index were the only significant positive predictors. Productivity exhibited a near one-to-one relationship with innovation. Automation, revenue growth, employee satisfaction, GDP growth, and inflation were not significant after controlling for AI adoption and productivity.

8. Revenue Growth Prediction Using Machine Learning

8.1 Train/Test Split

```r
set.seed(123)  # for reproducibility

trainIndex <- createDataPartition(
  data3$revenue_growth,
  p = 0.8,
  list = FALSE
)

train <- data3[trainIndex, ]
test <- data3[-trainIndex, ]
```

8.2 Random Forest Model

```r
# Train a random forest model with 500 trees
rf_model <- randomForest(
  revenue_growth ~
    ai_adoption_score +
    automation_level +
    productivity_index +
    innovation_score +
    employee_satisfaction +
    gdp_growth +
    inflation_rate,
  data = train,
  ntree = 500
)
```

8.3 Predictions and Model Comparison

```r
# Generate predictions on test set for both models
pred_rf <- predict(rf_model, newdata = test)
pred_lm <- predict(model_rev, newdata = test)
```

```r
# Compute RMSE for both models
rmse_rf <- rmse(test$revenue_growth, pred_rf)
rmse_lm <- rmse(test$revenue_growth, pred_lm)

rmse_rf
```

```
## [1] 10.38463
```

```r
rmse_lm
```

```
## [1] 10.29816
```

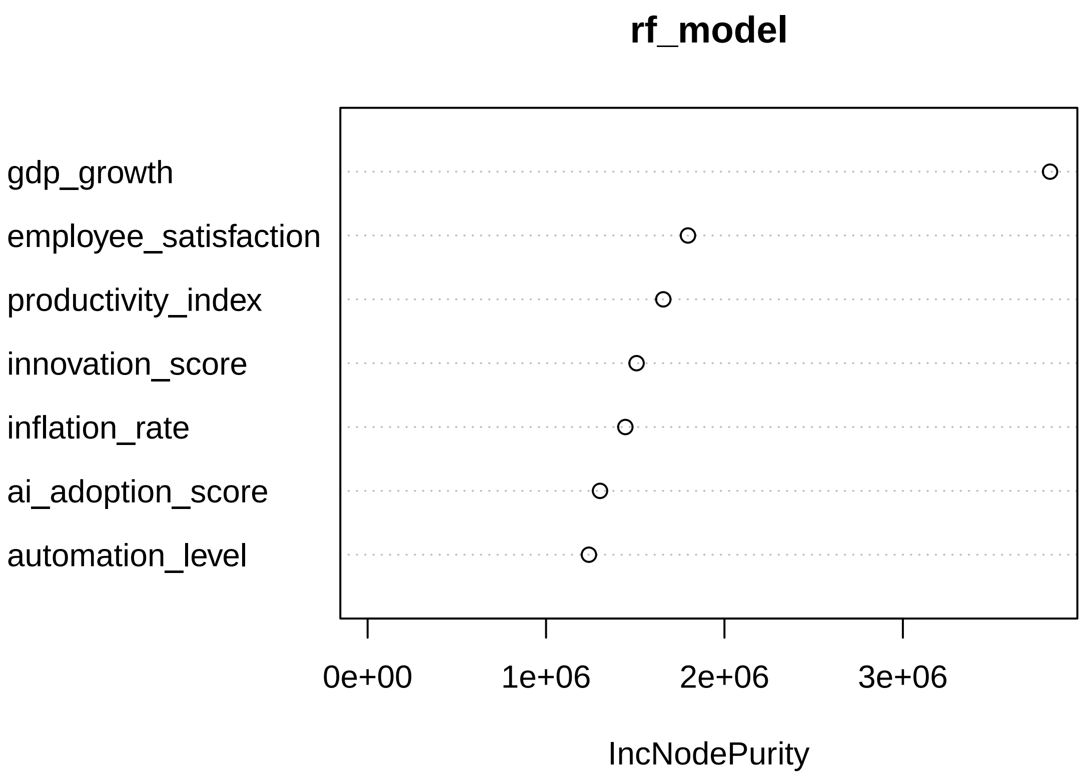
Interpretation: Linear regression performed very slightly better (lower RMSE) than the Random Forest. This suggests that the relationship between predictors and revenue growth is largely linear. The Random Forest's extra complexity did not improve prediction, making the linear model more parsimonious and interpretable.

8.4 Variable Importance (Random Forest)

```
# Display variable importance from the random forest
importance(rf_model)
```

```
##                        IncNodePurity
## ai_adoption_score            1302506
## automation_level             1240302
## productivity_index           1657246
## innovation_score             1507472
## employee_satisfaction        1795384
## gdp_growth                   3824898
## inflation_rate               1444526
```

```
varImpPlot(rf_model)
```



Random Forest Insights: GDP growth was the most influential predictor, followed by employee satisfaction and productivity index. AI adoption score also contributed meaningfully. Interestingly, innovation score showed high importance in the Random Forest despite being insignificant in the linear model, indicating potential nonlinear relationships. Automation level was the least important predictor.

```
# Compute standard regression performance metrics
postResample(pred = pred_rf, obs = test$revenue_growth)
```

```
##       RMSE   Rsquared        MAE
## 10.3846281  0.2414202  8.2914975
```

The Random Forest model achieved an R^2 of 0.241, with an RMSE of 10.38 and MAE of 8.29. This limited predictive power suggests that additional factors beyond those included may be required to fully explain revenue growth.

9. Summary and Next Steps

This analysis examined the simulated ai_workforce_transformation dataset, focusing on the Data Science department to understand how AI adoption correlates with organizational outcomes. Key findings include:

PCA confirmed that AI adoption clusters with innovation, productivity, and revenue growth, opposite to workforce instability indicators.

Linear models showed that AI adoption positively predicts revenue growth and innovation, but its association with layoff risk is comparatively modest.

Revenue growth is better explained by economic factors (GDP growth) and employee satisfaction than by automation alone.

Machine learning (Random Forest) did not outperform linear regression, indicating predominantly linear relationships in this synthetic dataset.

Further work could explore interaction effects, non-linear modeling for innovation, and subgroup analysis across industries. However, given the synthetic nature of the data, real-world validation would be essential for drawing actionable business conclusions.